

Supplementary Materials

Transfer learning from computed stability data for nanobody melting-temperature prediction

Taihei Murakami, Kentaro Sasaki, Soichiro Oda, Kazuma Okada, and Yasuhiro Matsunaga

Supplementary methods

Details of the heterogeneous MD data set

The heterogeneous data set was assembled from a SABDab summary and the corresponding structures [1]. For each PDB identifier, we selected the first heavy-chain entry whose chain was present in the structure file. This rule yielded 1143 samples with a usable sequence, topology, trajectory, and native-contact label, representing 833 unique amino-acid sequences. Ninety-seven sequences occurred in more than one PDB entry and accounted for 407 samples. We calculated native-contact values separately for each structure and sampled them individually during training; duplicate sequences were not collapsed, and the model did not receive the PDB identifier. Eleven sequences exceeded the 158-residue model input and were truncated at the end.

Structures were prepared and protonated at pH 7.4 with PDBFixer, pdb2pqr, and PROPKA [2, 3]. Each protein was solvated in OPC water with 15 Å padding and 0.15 M NaCl and described with ff19SB [4, 5]. Equilibration at 300 K used 1 ns NVT and 1 ns NPT with a 2-fs timestep, after which two 1-ns restrained NVT stages at 400 K preceded 100 ns of unrestrained 400-K NVT production. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps⁻¹ friction [6].

The contact calculation used all backbone heavy-atom pairs separated by more than three residues and within 0.45 nm in the first production frame, as for the single mutation scans. Q was averaged over the final 30 ns of production, that is the last 300 frames of 100-ps output. The two MD data sets thus shared one Q definition, contact set, native reference, and averaging window, and differed only in the variants they covered.

FEP charge-change check

For charge-changing mutations in a cubic 70-Å box and a solvent dielectric constant of 78, we tested the leading analytical periodic net-charge (Ewald self-energy) correction [7], $0.086(\Delta q)^2$ kcal mol⁻¹. We applied it and repeated the sign reversal, robust scaling, Yeo–Johnson transform with standardization, and min–max scaling used for the training labels. Across the 284 charge-changing mutations the scaled labels barely moved, with a maximum shift of 0.008 on the [0, 1] scale and a correlation of 0.99989 with the uncorrected labels, so the training labels use the uncorrected $\Delta\Delta G$.

Side-chain contacts in the native-contact label

The native-contact fraction Q in the main analysis uses backbone heavy atoms only. We also evaluated the side-chain-inclusive definition, in which all protein heavy atoms are eligible for contacts. On the 400 K single mutation scan this raised the label spread by roughly 1.4-fold, but it did not improve transfer. With hyperparameters held fixed it slightly increased the held-out T_m MAE, and re-selecting the learning rate, dropout, and encoder learning rate on the 114-example validation set did not recover the backbone result. In the fine-tuned regime the side-chain label was significantly worse than the T_m -only baseline (paired Δ MAE +0.30 °C, 95% CI [+0.12, +0.49]), while the backbone label was neutral; in the frozen regime it erased the backbone label’s benefit (Δ MAE from −0.20 to −0.04 °C). Because the mutant side chains were built without repacking their neighbors, side-chain contacts near each mutation carry construction noise rather than added signal, so the main analysis retains the backbone-only definition.

Model-size checks

We fine-tuned the 35M- and 650M-parameter ESM2 encoders without size-specific hyperparameter searches. For both sizes, the T_m -only model used the shared architecture, dropout 0.05, and an encoder learning rate of 1×10^{-4} . The FEP model used the shared architecture with separate 1MEL and 4IDL heads, $n = 320$ samples per structure, dropout 0.15, and an encoder learning rate of 3×10^{-5} . Both conditions used a non-encoder learning rate of 3×10^{-4} , weight decay 0.04, and ten training seeds.

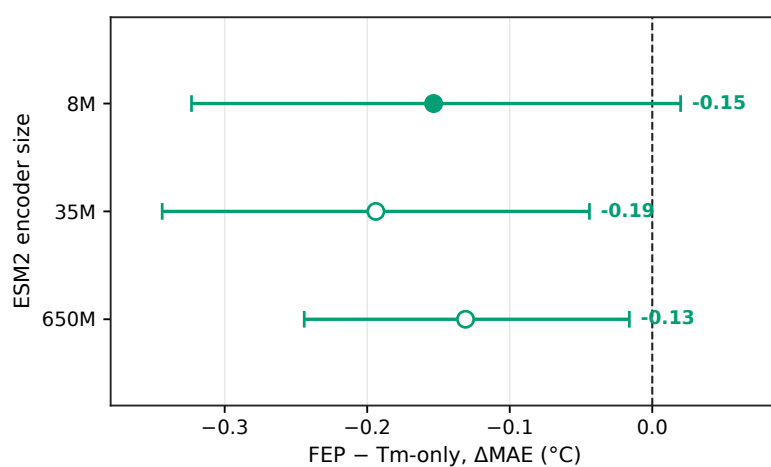


Figure S1 The FEP result held across ESM2 encoder sizes. FEP-minus-Tm-only MAE (ΔMAE) with a fine-tuned encoder; negative values mean lower Tm error with FEP. The 8M point is the five-seed model selected by the two-stage search, and the 35M and 650M points are ten-seed checks without size-specific tuning. Whiskers are 95% paired bootstrap intervals over the 396 Tm test sequences.

References

- [1] Dunbar, J., Krawczyk, K., Leem, J., Baker, T., Fuchs, A., Georges, G., et al. SAbDab: the structural antibody database. *Nucleic Acids Research* 42, D1140–D1146 (2014). <https://doi.org/10.1093/nar/gkt1043>
- [2] Dolinsky, T. J., Nielsen, J. E., McCammon, J. A., Baker, N. A. PDB2PQR: an automated pipeline for the setup of Poisson–Boltzmann electrostatics calculations. *Nucleic Acids Research* 32, W665–W667 (2004). <https://doi.org/10.1093/nar/gkh381>
- [3] Olsson, M. H. M., Søndergaard, C. R., Rostkowski, M., Jensen, J. H. PROPKA3: consistent treatment of internal and surface residues in empirical pK_a predictions. *Journal of Chemical Theory and Computation* 7, 525–537 (2011). <https://doi.org/10.1021/ct100578z>
- [4] Tian, C., Kasavajhala, K., Belfon, K. A. A., Raguet, L., Huang, H., Miguels, A. N., et al. ff19SB: amino-acid-specific protein backbone parameters trained against quantum mechanics energy surfaces in solution. *Journal of Chemical Theory and Computation* 16, 528–552 (2020). <https://doi.org/10.1021/acs.jctc.9b00591>
- [5] Izadi, S., Anandakrishnan, R., Onufriev, A. V. Building water models: a different approach. *The Journal of Physical Chemistry Letters* 5, 3863–3871 (2014). <https://doi.org/10.1021/jz501780a>
- [6] Eastman, P., Swails, J., Chodera, J. D., McGibbon, R. T., Zhao, Y., Beauchamp, K. A., et al. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology* 13, e1005659 (2017). <https://doi.org/10.1371/journal.pcbi.1005659>
- [7] Rocklin, G. J., Mobley, D. L., Dill, K. A., Hünenberger, P. H. Calculating the binding free energies of charged species based on explicit-solvent simulations employing lattice-sum methods: An accurate correction scheme for electrostatic finite-size effects. *The Journal of Chemical Physics* 139, 184103 (2013). <https://doi.org/10.1063/1.4826261>